

SEMANTICS OF HOMOTOPY TYPE THEORY

STEFANO LUNEIA

CATEGORICAL SEMANTICS FOR TYPE THEORY

. STAC (SIMPLY TYPED)

Consider a CARTESIAN CLOSED CATEGORY, $\mathcal{C}$

then

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ASSUME UNIT
TYPE, PRODUCT
AND FUNCTION
TYPE

$$* \rightsquigarrow \llbracket * \rrbracket = 1 \quad \text{TERMINAL OBJECT}$$

$$T \rightsquigarrow \llbracket T \rrbracket \quad \text{OBJECT OF } \mathcal{C}$$

$$T \times S \rightsquigarrow \llbracket T \times S \rrbracket = \llbracket T \rrbracket \times \llbracket S \rrbracket$$

$$S \rightarrow T \rightsquigarrow \llbracket S \rightarrow T \rrbracket = \llbracket T \rrbracket^{\llbracket S \rrbracket}$$

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. STAC (SIMPLY TYPED)

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$$[[*]] = 1$$

$$[[T]] \text{ OBJECT of } \mathcal{C}$$

$$[[T \times S]] = [[T]] \times [[S]]$$

$$[[S \rightarrow T]] = [[T]]^{[[S]]}$$

SOUNDNESS: if premises
are verified in $\mathcal{C}$ so
is their conclusion

↓

CCC are SOUND MODELS
for STAC

• MLDTT (DEPENDENT TYPE THEORY)

To be able to interpret DEPENDENT TYPES
we need more structure than CCC



LOCALLY CARTESIAN CLOSED CATEGORY (LCCC)
ARE SOUND MODELS FOR MLDTT

• SYNTACTIC CATEGORY or CATEGORY OF CONTEXT

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• MORPHISMS: TERMS i.e. $f: A \rightarrow B$ given by $x:A \vdash f:B$

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- IDENTITY

$$x:A \vdash x:A$$

VARIABLE RULE

- COMPOSITION

$$\frac{y:B \vdash g:C}{x:A, y:B \vdash g:C}$$

$$\frac{x:A, y:B \vdash g:C}{x:A \vdash g[f/y]:C}$$

SUBSTITUTION

$$f:A \rightarrow B \quad g:B \rightarrow C$$

$$x:A \vdash g[f/y]:C$$

More precisely $\text{Syn}(\Pi)$

• OBJECTS: CONTEXTS Γ, Δ

$$\Gamma = (x_1 : \Gamma_1, \dots, x_n : \Gamma_n)$$

• MORPHISMS: CONTEXTS MORPHISMS

$$\Gamma = (x_1 : \Gamma_1, \dots, x_n : \Gamma_n)$$

$$\Gamma \vdash t_1(x_0, \dots, x_n) : \Delta_0$$

$$\vdots$$

$$\Gamma \vdash t_m(x_0, \dots, x_n) : \Delta_m$$

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— IDENTITIES

$$\text{id}_\Gamma = (x_1, \dots, x_n)$$

- COMPOSITION

$$\Gamma = (x_1: \Gamma_1, \dots, x_n: \Gamma_n)$$

$$\downarrow \sigma = (s_1, \dots, s_m)$$

$$\Gamma \vdash s_i(x_1, \dots, x_n): \Delta_i$$

$$\Delta = (\gamma_1: \Delta_1, \dots, \gamma_m: \Delta_m)$$

$$\downarrow \tau = (t_1, \dots, t_\kappa)$$

$$\Delta \vdash t_i(\gamma_1, \dots, \gamma_m): H_i$$

$$\vdash = (z_1: H_1, \dots, z_\kappa: H_\kappa)$$

$$\tau \sigma = \Gamma \vdash t_i \left[\frac{s_j}{\gamma_i} \right]: H_i$$

||

$$t_i(s_1(x_1, \dots, x_n), \dots, s_m(x_1, \dots, x_n))$$

NOTE : Latter definition of $\text{Syn}(\Pi)$ is "THE CORRECT ONE", but as long Π HAS PRODUCT THE TWO DEFINITION ARE EQUIVALENT

Otherwise we could describe terms like
 $x:A, y:B \vdash z:C$ wannabe " $A \times B \rightarrow C$ "

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To have INTENSIONAL TYPE THEORY WE NEED HIGHER
HOMOTOPICAL STRUCTURE $\rightsquigarrow$ VARIOUS WAY OF DOING
THAT (DIFFERENT MODELS)

DISPLAY MAP CATEGORIES

Def. $(\mathcal{C}, D)$ is a DISPLAY MAP CATEGORY, with $D \subset \text{Mor}(\mathcal{C})$

- if
- $\mathcal{C}$ HAS A TERMINAL OBJECT $*$
 - D IS CLOSED UNDER PULLBACKS

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INTERPRETATION

CONTEXTS



OBJECTS Γ of $\mathcal{C}$

DEP. TYPES



DISPLAY MAPS $\Gamma.A \twoheadrightarrow \Gamma$

$(\Gamma \vdash A \text{ type})$

TERMS



SECTION OF DISPLAY MAPS



UNIT TYPE

unit



*

$\vdash \text{unit TYPE}$

unit



$\vdash t : \text{unit}$

$x : \text{unit} \vdash D(x) \text{ TYPE}$

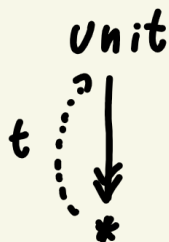
$\Gamma \vdash d : D(t)$

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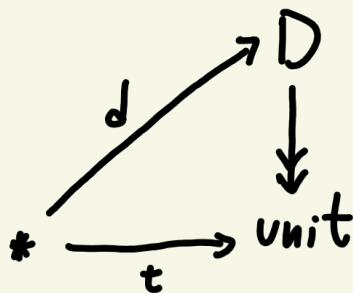
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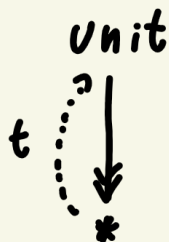
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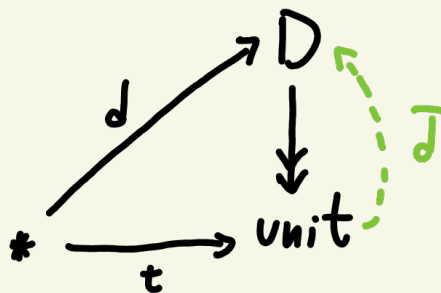
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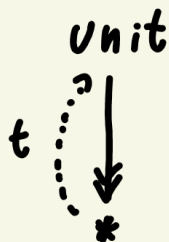
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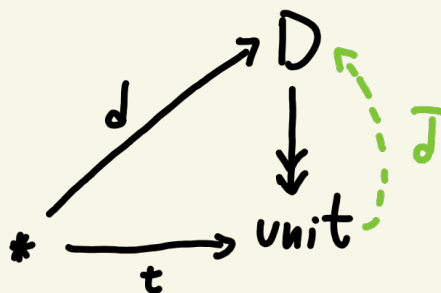
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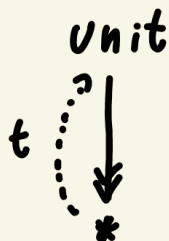
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So (C, D) MODELS THE UNIT TYPE

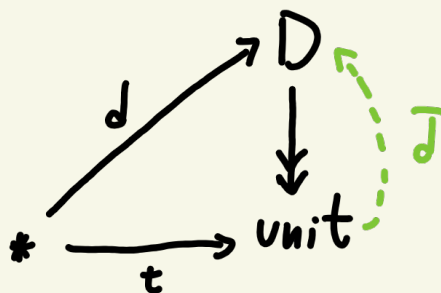
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So (C, D) MODELS THE UNIT TYPE

if $* \xrightarrow{\text{id}_*} * \in D$

This also forces D to be closed under ISOMORPHISMS

Σ - TYPES

$$AB=B \longrightarrow A \longrightarrow *$$

$$\sum_A B \dashrightarrow *$$

$$\frac{x:A \vdash B(x) \text{ TYPE}}{\vdash \sum_{x:A} B(x) \text{ TYPE}}$$

$$\frac{x:A, y:B(x)}{\langle x, y \rangle : \sum_{x:A} B(x)}$$

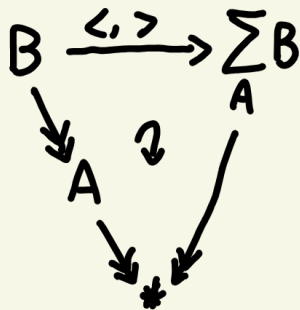
$$\frac{z: \sum_{x:A} B(x) \vdash D(z) \text{ TYPE} \quad x:A, y:B(x) \vdash d: D(\langle x, y \rangle)}{z: \sum_{x:A} B(x) \vdash \text{ind}_{\Sigma}(d, z): D(z)}$$

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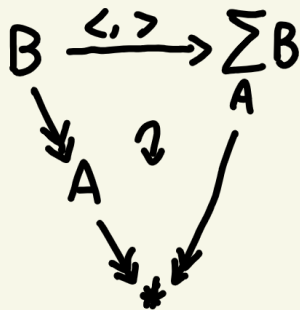
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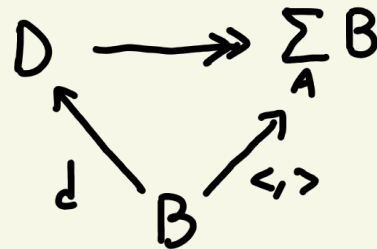
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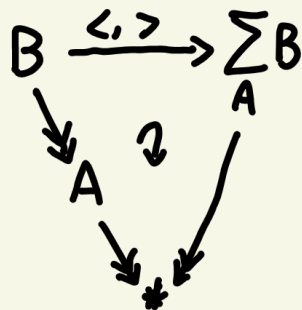
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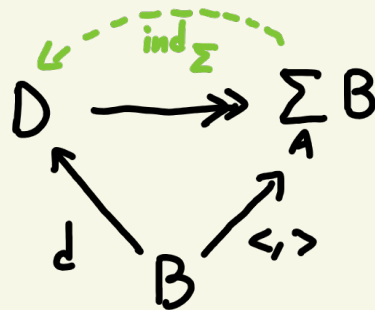
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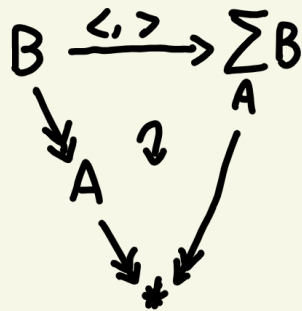
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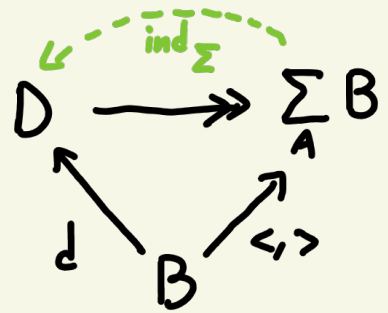
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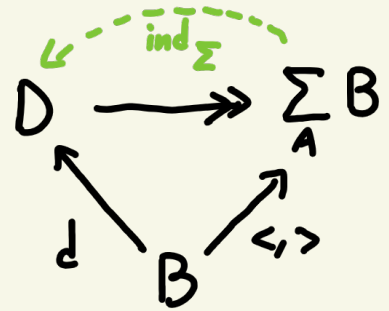
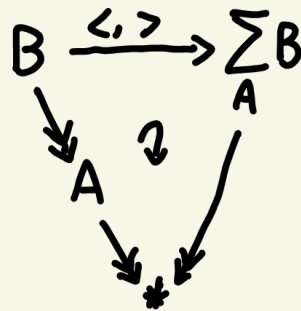
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So (C, D) MODELS Σ -TYPES

if D is CLOSED under COMPOSITION

$\sum_A B := B$ (projection on B is then an iso as prescribed by computation)

Π -TYPES

$$B \longrightarrow A \longrightarrow *$$

$$\prod_A B \dashrightarrow *$$

$$\frac{x:A \vdash B(x) \text{ TYPE}}{\vdash \prod_{x:A} B(x) \text{ TYPE}}$$

$$\frac{x:A \vdash b(x) : B(x)}{\vdash \lambda x. bx : \prod_{x:A} B(x)}$$

$$\frac{\Gamma \vdash f : \prod_{x:A} B(x)}{\Gamma, x:A \vdash f(x) : B(x)}$$

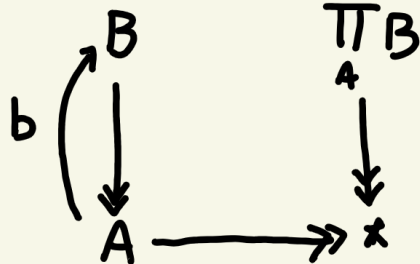
Π -TYPES

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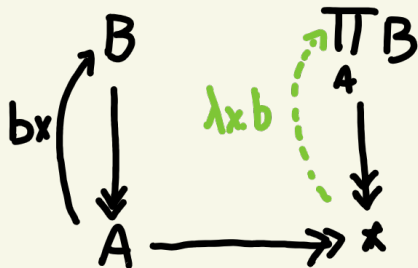
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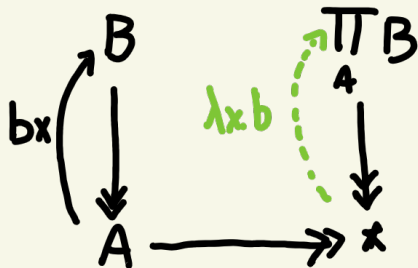
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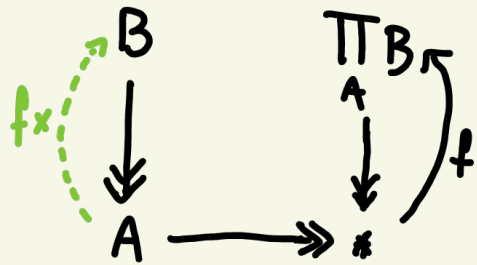
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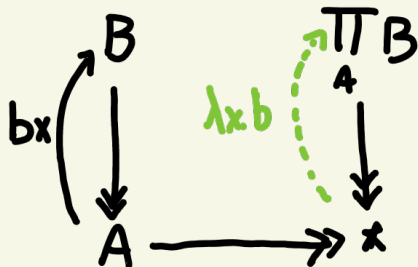
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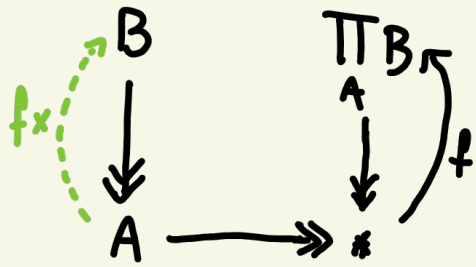
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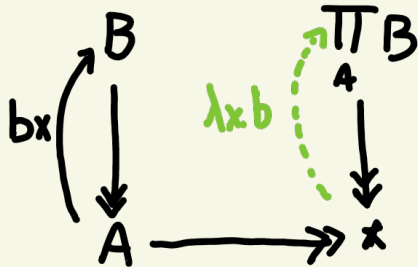
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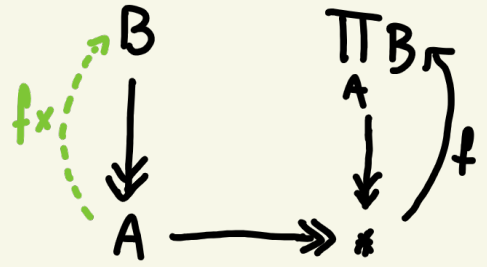
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if $\mathcal{C}$ is LCC and

if $f \in D$ then $\prod_g f \in D$

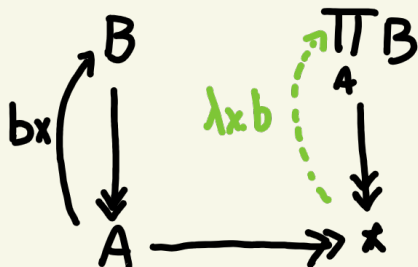
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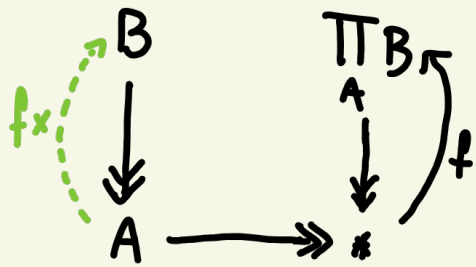
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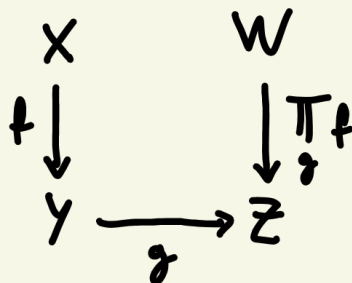
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if $\mathcal{C}$ is LCC and

i.e.

if $f \in D$ then $\prod_y f \in D$



$$\text{s.t. } \text{hom}_{\mathcal{C}/Z}(k, \prod_y f) \cong \text{hom}_{\mathcal{C}/Y}(g^* k, f)$$

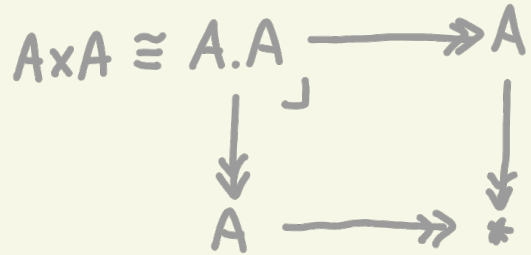
Id - TYPES

Id_A



$A \times A$

where



$a, b: A \vdash \text{Id}_A(a, b) \text{ TYPE}$

$a: A \vdash r_a: \text{Id}_A(a, a)$

Id - TYPES

$$\begin{array}{c} \text{Id}_A \\ \downarrow \\ A \times A \end{array}$$

$$a, b: A \vdash \text{Id}_A(a, b) \text{ TYPE}$$

$$\begin{array}{ccc} A \times A \cong A.A & \xrightarrow{\quad} & A \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow{\quad} & * \end{array}$$

where

$$\begin{array}{ccc} \text{Id}_A.\Delta & \xrightarrow{\quad} & \text{Id}_A \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow[\Delta]{} & A \times A \end{array}$$

$$a: A \vdash r_2: \text{Id}_A(a, a)$$

Id - TYPES

$$\begin{array}{c} \text{Id}_A \\ \downarrow \\ A \times A \end{array}$$

where

$$\begin{array}{ccc} A \times A \cong A.A & \xrightarrow{\quad} & A \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow{\quad} & * \end{array}$$

$$a, b: A \vdash \text{Id}_A(a, b) \text{ TYPE}$$

$$\begin{array}{ccc} \text{Id}_A.\Delta & \xrightarrow{\quad} & \text{Id}_A \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow{\Delta} & A \times A \end{array}$$

(Note: Dashed green arrows labeled r_A point from A to $\text{Id}_A.\Delta$ and from A to Id_A)

$$a: A \vdash r_a: \text{Id}_A(a, a)$$

N.B. Asking for the existence of the lift is the same as asking for the section

Id - TYPES

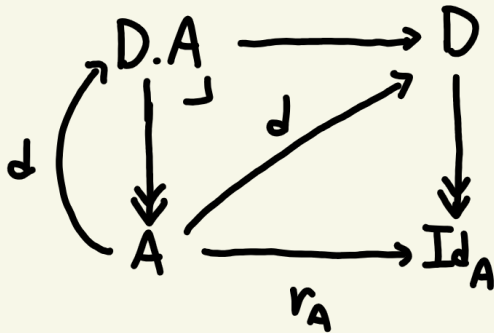
$$a, b: A, p: \text{Id}_A(a, b) \vdash D(a, b, p)$$

$$a: A \vdash d: D(a, b, r_A)$$

elim and
comp

$$\frac{}{a, b: A, p: \text{Id}_A(a, b) \vdash \bar{d}(a, b, p): D(a, b, p)}$$

$$a: A \vdash da \doteq \bar{d}(a, a, r_A): D(a, a, r_A)$$



Id - TYPES

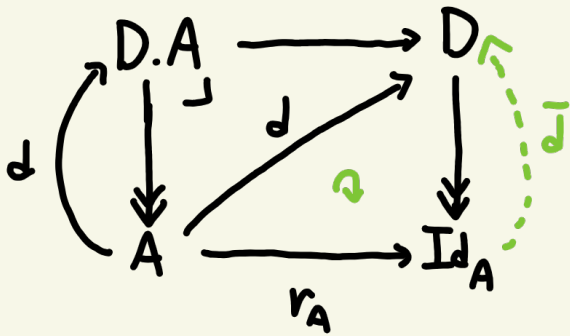
$$a, b: A, p: \text{Id}_A(a, b) \vdash D(a, b, p)$$

$$a: A \vdash d: D(a, b, r_A)$$

elim and
comp

$$\frac{}{a, b: A, p: \text{Id}_A(a, b) \vdash \bar{d}(a, b, p): D(a, b, p)}$$

$$a: A \vdash da \doteq \bar{d}(a, a, r_A): D(a, a, r_A)$$



Id - TYPES

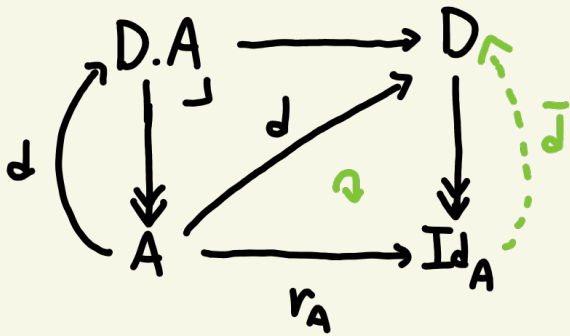
$$a, b: A, p: \text{Id}_A(a, b) \vdash D(a, b, p)$$

$$a: A \vdash d: D(a, b, r_A)$$

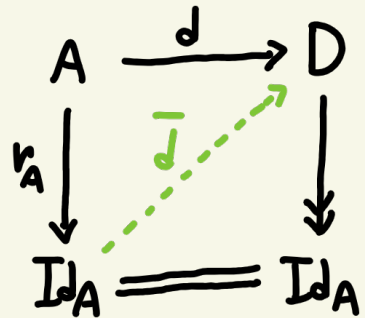
elim and
comp

$$\frac{}{a, b: A, p: \text{Id}_A(a, b) \vdash \bar{d}(a, b, p): D(a, b, p)}$$

$$a: A \vdash da \doteq \bar{d}(a, a, r_A): D(a, a, r_A)$$



equivalently



$\bar{d}$ SOLUTION TO THE
LIFTING PROBLEM

Id - TYPES

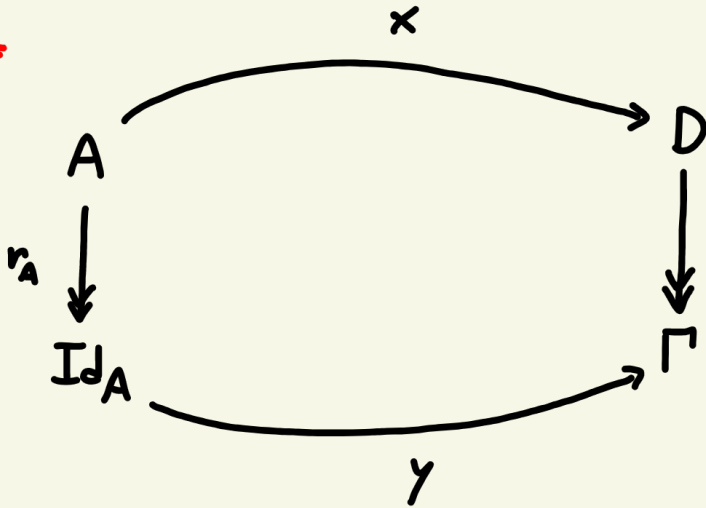
Prop. $\{r_A \mid A \in \mathcal{C}\} \not\sqsubseteq D$

PROOF

Id - TYPES

Prop. $\{r_A \mid A \in \mathcal{C}\} \not\sqsubseteq D$

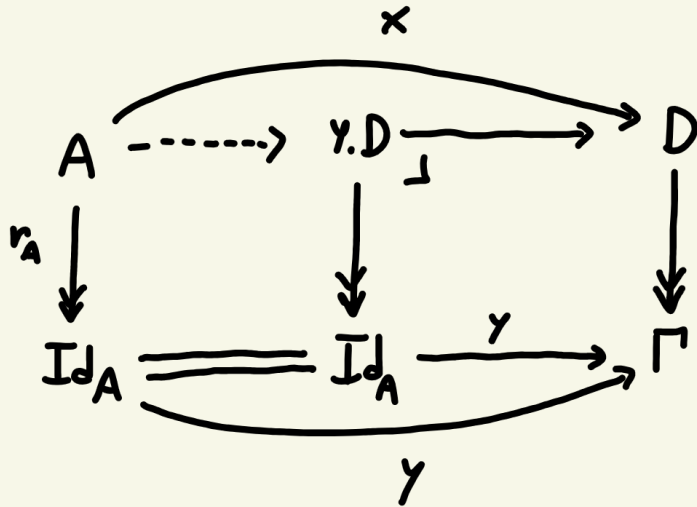
PROOF



Id - TYPES

Prop. $\{r_A \mid A \in \mathcal{C}\} \not\sqsubseteq D$

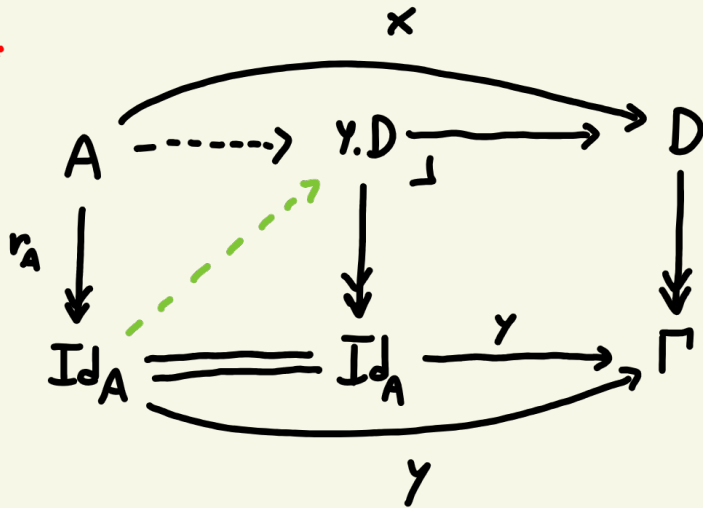
PROOF



Id - TYPES

Prop. $\{r_A \mid A \in \mathcal{C}\} \not\sqsubseteq D$

PROOF

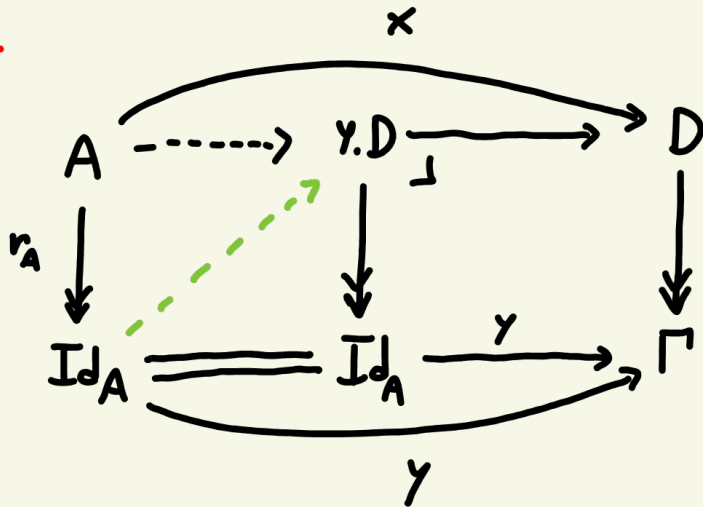


□

Id - TYPES

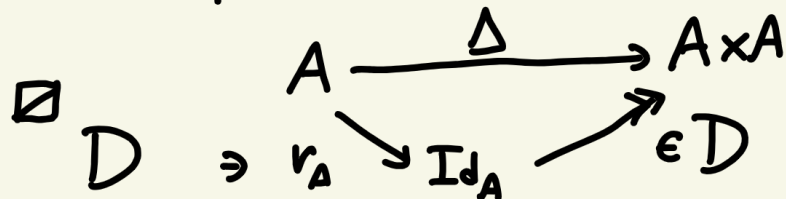
Prop. $\{r_A \mid A \in \mathcal{C}\} \not\subseteq \mathcal{D}$

PROOF



□

Note: we get the factorization



Id - TYPES

So (C, D) MODELS Id - TYPES

Id-TYPES

So $(\mathcal{C}, D)$ MODELS Id-TYPES

it $(I, R) := (\Box D, (\Box D)^\Box)$ IS A W.F.S. s.t.

I IS STABLE UNDER PULLBACK ALONG R

Id - TYPES

So $(\mathcal{C}, D)$ MODELS Id - TYPES

it $(\mathcal{I}, \mathcal{R}) := (\Box D, (\Box D)^\Box)$ is a w.f.s. s.t.

$\mathcal{I}$ IS STABLE UNDER PULLBACK ALONG $\mathcal{R}$

Moreover for any map $X \xrightarrow{f} Y$ in $\mathcal{C}$ the factorization is given by

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow X \times_{\eta_f} & \nearrow \varepsilon_1 \circ \pi_{\text{Id}_Y} \\ & X \times \text{Id}_Y & \end{array}$$

Mapping Path Space

Thank you!